

Title

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EXECUTIVE SUMMARY



Analysis of the 2025 Stack Overflow Developer Survey and job posting data reveals clear technology trends for data professionals.

- Python, JavaScript, and SQL remain the most in-demand programming languages in job postings.
- Washington DC (5,316) and Detroit (3,945) lead U.S. cities in data analyst job postings.
 - PostgreSQL and MongoDB are gaining ground on traditional relational databases.
 - MySQL and PostgreSQL dominate future database preference among developers.
 - Survey demographics show a predominantly male workforce; gender diversity remains a key challenge.
- Python job postings number 1,173, confirming its critical role in the data science ecosystem.
- Cloud and web technologies (JavaScript, Java) are increasingly required alongside traditional analytics skills.
- Organizations should invest in upskilling teams in Python, cloud databases, and modern BI tools.



INTRODUCTION



Purpose, Target Audience & Value of the Analysis

- This capstone analyzes real-world developer survey data and job postings to identify technology hiring trends for 2025-2026.
- Target audience: HR professionals, business leaders, and aspiring data analysts seeking market intelligence.
- The analysis covers programming languages, database technologies, web frameworks, and workforce demographics.
- Data sources include the Stack Overflow Developer Survey (2025) and a real-time jobs API with thousands of postings.
 - Findings enable data-driven decisions on training investments and recruitment strategies.
 - Results are presented through interactive dashboards and visualizations for easy interpretation.



METHODOLOGY

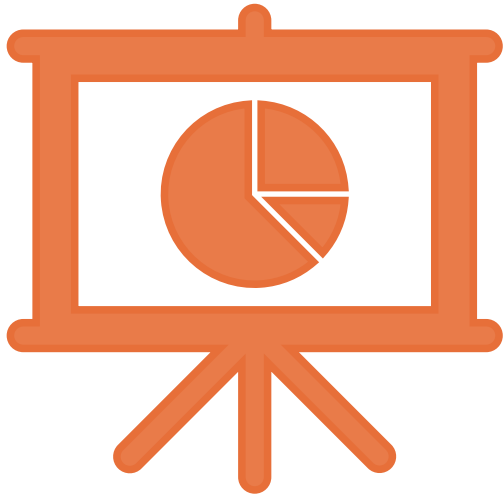


Data Sources, Collection & Wrangling Steps

- Job postings collected via IBM Jobs API (jobs.json) covering 7 major U.S. cities and 12 key technologies.
- Language salary data collected via web scraping of developer salary survey pages into popular-languages.csv.
- Technology trend data sourced from the Stack Overflow Developer Survey 2025 (current year) and future preference responses.
- Data wrangling steps included deduplication, null handling, normalization, and export to Excel/CSV formats.
 - Python (pandas, openpyxl, requests) used for all data collection, cleaning, and analysis tasks.
 - Dashboards built in IBM Cognos Analytics using cleaned datasets for interactive visual exploration.



Results



<Please present your results in the following slides.>



JOB POSTINGS

Job postings by location (descending order). Washington DC leads with 5,316 postings, followed by Detroit (3,945), Seattle (3,375), New York (3,226), Los Angeles (640), San Francisco (435), and Austin (434). The data was collected via the IBM Jobs API from jobs.json.



POPULAR LANGUAGES

Top programming languages by average annual salary (USD). Scala leads at \$117,313, followed by Go (\$92,773), Oracle (\$97,037), Swift (\$92,592), and Objective-C (\$85,656). Python averages \$79,578. Data collected via web scraping into popular-languages.csv.

PROGRAMMING LANGUAGE TRENDS

Python is the fastest-growing language, rising from 4th to 3rd place and gaining 5.3 percentage points. TypeScript also shows strong growth (+4.3%), while Java and PHP are declining. JavaScript and HTML/CSS maintain dominance across both years.

Current Year

Top 10 (Current): JavaScript (67.7%), HTML/CSS (63.1%), SQL (54.7%), Python (44.1%), TypeScript (38.9%), Java (35.4%), C# (27.4%), Bash (27.2%), C++ (23.9%), PHP (21.5%)

Next Year

Top 10 (Next Year): JavaScript (63.5%), HTML/CSS (57.4%), Python (49.4%), SQL (47.1%), TypeScript (43.2%), Java (29.5%), C++ (24.4%), C# (22.8%), Go (18.0%), Kotlin (15.9%)



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Python is rapidly gaining popularity and is projected to overtake SQL in future developer usage, reflecting its importance in data science and automation.
- TypeScript continues to grow at the expense of plain JavaScript, signaling a shift toward typed, scalable web development.
- Legacy languages (PHP, C#) show declining trend, suggesting organizations are modernizing their tech stacks.

Implications

- Training programs should prioritize Python and TypeScript to align with projected market demand.
- Data teams should ensure Python proficiency as a baseline skill given its cross-domain applications.
- Organizations using legacy PHP or C# stacks should plan migration timelines to stay competitive in talent acquisition.



DATABASE TRENDS

PostgreSQL overtakes MySQL as the most desired database, rising from 2nd to 1st place. MongoDB and Redis are gaining significantly. Microsoft SQL Server drops sharply in future preferences, suggesting a shift toward open-source database solutions.

Current Year

Top 10 Databases (Current): MySQL (55.6%), PostgreSQL (47.3%), SQLite (43.4%), MongoDB (28.3%), Microsoft SQL Server (26.7%), Redis (18.1%), MariaDB (17.4%), Elasticsearch (12.2%), Oracle (11.3%), DynamoDB (9.7%)

Next Year

Top 10 Databases (Next Year): PostgreSQL (41.1%), MySQL (32.7%), MongoDB (22.7%), Redis (18.4%), SQLite (16.6%), Elasticsearch (11.2%), DynamoDB (10.8%), MariaDB (9.6%), Microsoft SQL Server (9.0%), Firebase (8.9%)



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is becoming the preferred relational database, surpassing MySQL in future developer preferences for the first time.
- NoSQL databases (MongoDB, Redis, DynamoDB) are growing in popularity, reflecting the rise of unstructured and real-time data workloads.
- Microsoft SQL Server usage is projected to decline significantly, as developers favor open-source alternatives.

Implications

- Enterprises should evaluate migrating from SQL Server to PostgreSQL to align with future talent supply and reduce licensing costs.
- Data engineering teams should build competency in both relational (PostgreSQL) and NoSQL (MongoDB, Redis) databases.
- Cloud-native databases (DynamoDB, Firebase) signal growing adoption of serverless architectures that hiring plans should account for.

DASHBOARD

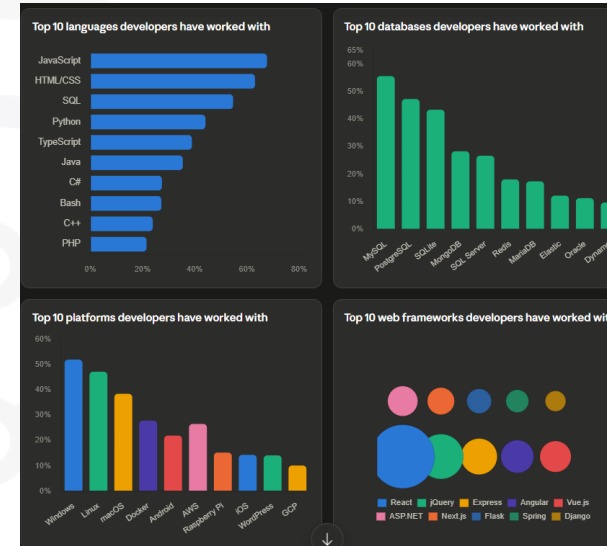


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DASHBOARD TAB 1: Current Technology Usage

- Dashboard Insight
- JavaScript dominates current usage (67.7%), followed by HTML/CSS and SQL. Python ranks 4th, showing strong adoption. MySQL leads database usage at 55.6%. Windows and Linux are the most used operating systems among developers.

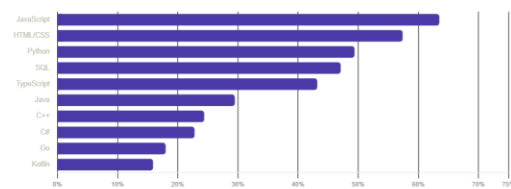


DASHBOARD TAB 2: Future Technology Trends

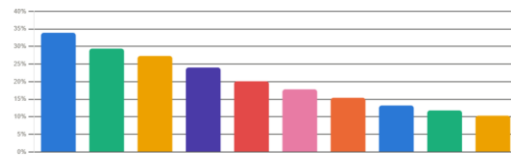
- Dashboard Insight
- Python is expected to surpass SQL in future usage. TypeScript and Go show the largest growth. PostgreSQL overtakes MySQL as the most desired database. React and Node.js remain top web frameworks developers plan to adopt next year.

IBM COGNOS ANALYTICS
Future Technology Trend

Top 10 languages developers want to work with next year

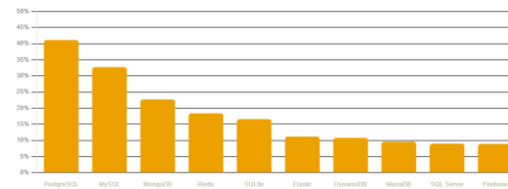


Top 10 platforms developers want to work with next year

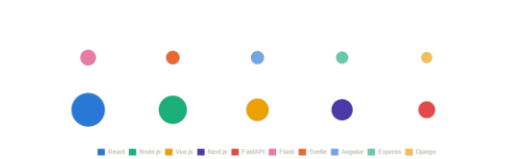


2025 Survey Next Year Preferences

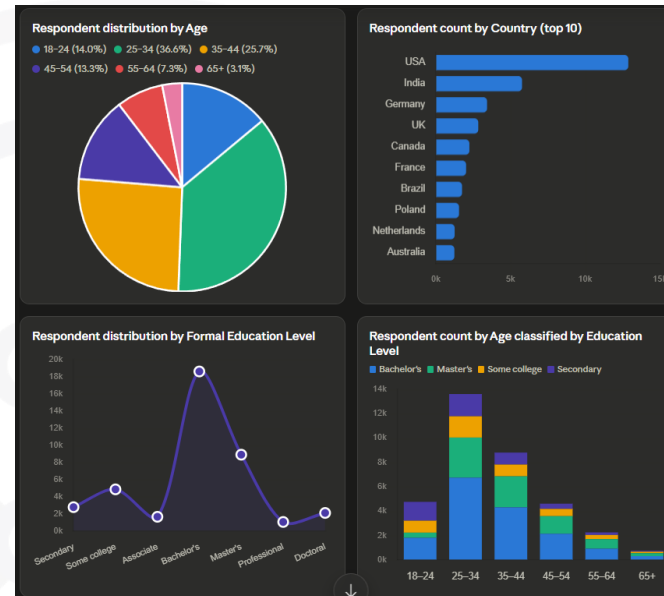
Top 10 databases developers want to work with next year



Top 10 web frameworks developers want to work with next year



DASHBOARD TAB 3: Demographics



- Dashboard Insight
- The respondent population is predominantly male (~91%). The USA has the highest number of respondents, followed by India and Germany. Most developers are aged 25-34. A significant majority hold a bachelor's degree or higher. Remote and hybrid work preferences dominate post-pandemic.

DISCUSSION



- The dashboards collectively reveal a clear shift toward open-source, cloud-native, and Python-centric technology stacks. Current usage data confirms JavaScript and MySQL dominance, while future preference data shows Python and PostgreSQL rising sharply. Demographic data highlights an opportunity for greater diversity in tech hiring. The job posting analysis confirms that organizations in major U.S. cities are actively seeking professionals with Python, Java, and Oracle skills, with Washington DC and Detroit showing the highest demand. These trends suggest the industry is evolving toward more versatile, scalable, and cost-efficient technology choices.



OVERALL FINDINGS & IMPLICATIONS

Findings

- Python is the single most important skill for data professionals, appearing in job postings (1,173) and ranking as the fastest-growing programming language in developer surveys.
- The tech industry is undergoing a database shift from proprietary (SQL Server, Oracle) to open-source solutions (PostgreSQL, MongoDB, Redis).
- Geographic job concentration is high: Washington DC, Detroit, and Seattle account for over 70% of the analyzed U.S. data analyst job postings.

Implications

- Workforce development programs should make Python and SQL core competencies, supplemented by cloud and NoSQL database skills.
- Technology leaders should accelerate open-source database adoption to reduce vendor dependency and attract modern talent pools.
- Companies outside top-3 cities (DC, Detroit, Seattle) should invest in remote-work policies to access a broader pool of data talent.



CONCLUSION



- This analysis confirms that Python, JavaScript, and SQL are essential skills for any data professional entering today's job market.
- The database landscape is shifting toward open-source (PostgreSQL, MongoDB), and organizations should proactively upskill teams accordingly.
- Job markets are geographically concentrated, but remote work trends present opportunities for talent democratization across regions.
- Diversity remains an industry challenge; future surveys and hiring pipelines should prioritize inclusion to unlock broader innovation potential.



APPENDIX



- Include any relevant additional charts, or tables that you may have created during the analysis phase.

